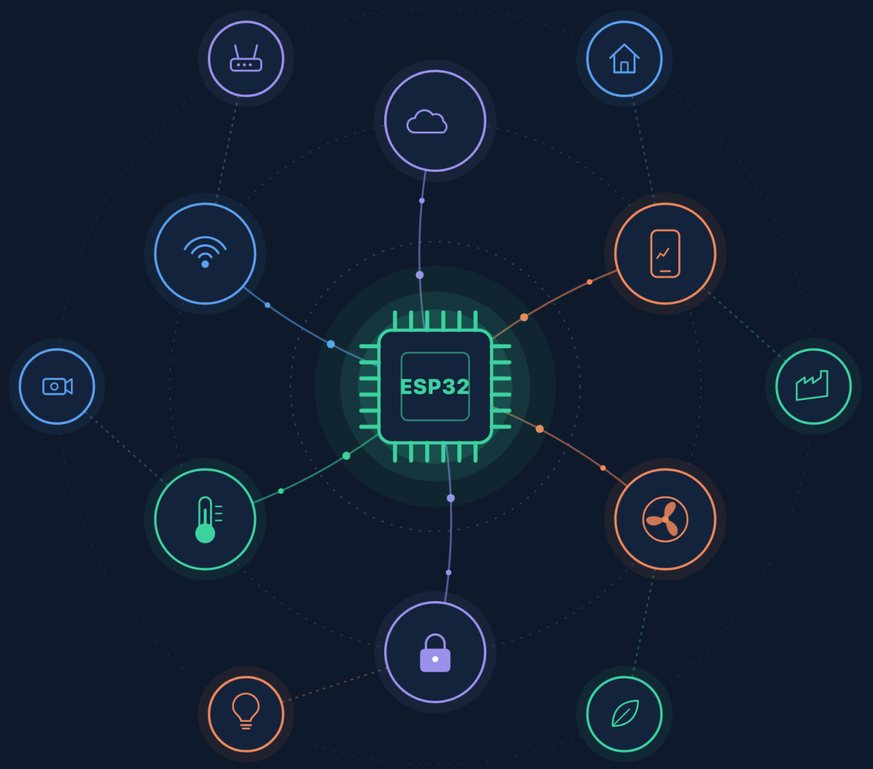
MODULE 5 · 45 MINUTES

IoT Cloud Platforms and Data Visualisation

Where a tower site’s readings go, and how to build dashboards for the NOC and for managers.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
Lecture

02
How it's built
Lecture

03
Sensors and actuators
Lab 2

04
Getting online
Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05
Cloud and dashboards
You are here

06
Automation
Lab 5

07
Security
Group activity

08
Real world + pitch
Lab 7

BY THE END OF THIS MODULE

You will be able to...

01 **Explain** what an IoT cloud platform does with site data

02 **Design** two views: one for the NOC, one for managers

03 **Choose** the right chart for each question

04 **Describe** an energy view: diesel, energy mix and PUE

05 **Build** a live trend view of a site’s shelter in Lab 4

Part A

From the tower site to the cloud

In Lab 3, your readings vanished when you closed the tab. Where should they live?

FROM THE SITE TO THE DASHBOARD

What an IoT platform does

Ingest

Readings arrive over the network.

Store

Every reading is kept with its time.

Analyse

Averages, trends and alerts.

Visualise

Charts and dashboards for people.

Rules and actions

“If this, then that”: in Module 6.

The cloud means renting computers and storage over the internet. It runs in data centres.

THE PLATFORM FOR LAB 4

ThingSpeak: channels, fields and keys

Channels

One per project or device, like “Tower Site 12”.

Fields

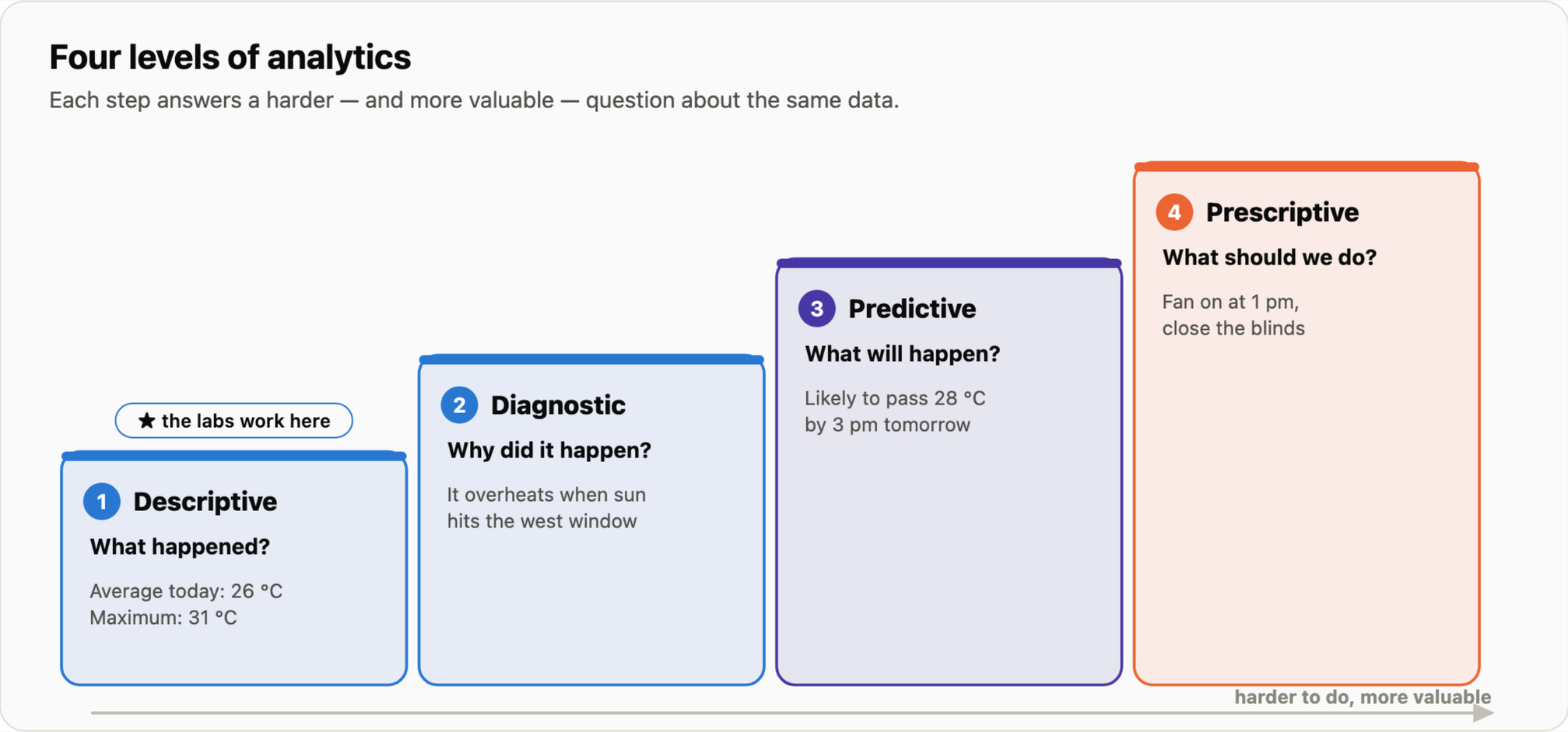
Up to 8 values per channel: Field 1 = Temperature.

API keys

The **Write** key adds data. The **Read** key reads a private channel.

Free plan: about 3 million messages a year, 4 channels, **15 s** between updates, non-commercial

Four kinds of analytics



Part B

Dashboards for two audiences

One set of data, two very different screens.



Photo: pslim, CC BY 2.0, via Wikimedia Commons

SMART TOWER MONITORING

One NOC screen, many sites

- **One screen** for hundreds of tower sites
- **A status light per site:** OK, warning or alarm
- **Alarms now:** shelter hot, door open, generator on
- **Worst first**, with a word, not only a colour
- **Action:** call the site, or send an engineer

SAME DATA, TWO SCREENS

Two dashboards for two audiences

	Operations (NOC)	Management
Question	Is any site in trouble **now**?	Are we getting **better**?
Time range	The last hour	Last month, last year
Headline	Alarms: shelter hot, door open	Energy cost, diesel, uptime
Best visual	Status lights and gauges	Big numbers and line charts
Action	Send a field engineer	Approve a budget, set a target

Rule one of every dashboard: **know your audience**.

CHOOSING THE RIGHT CHART

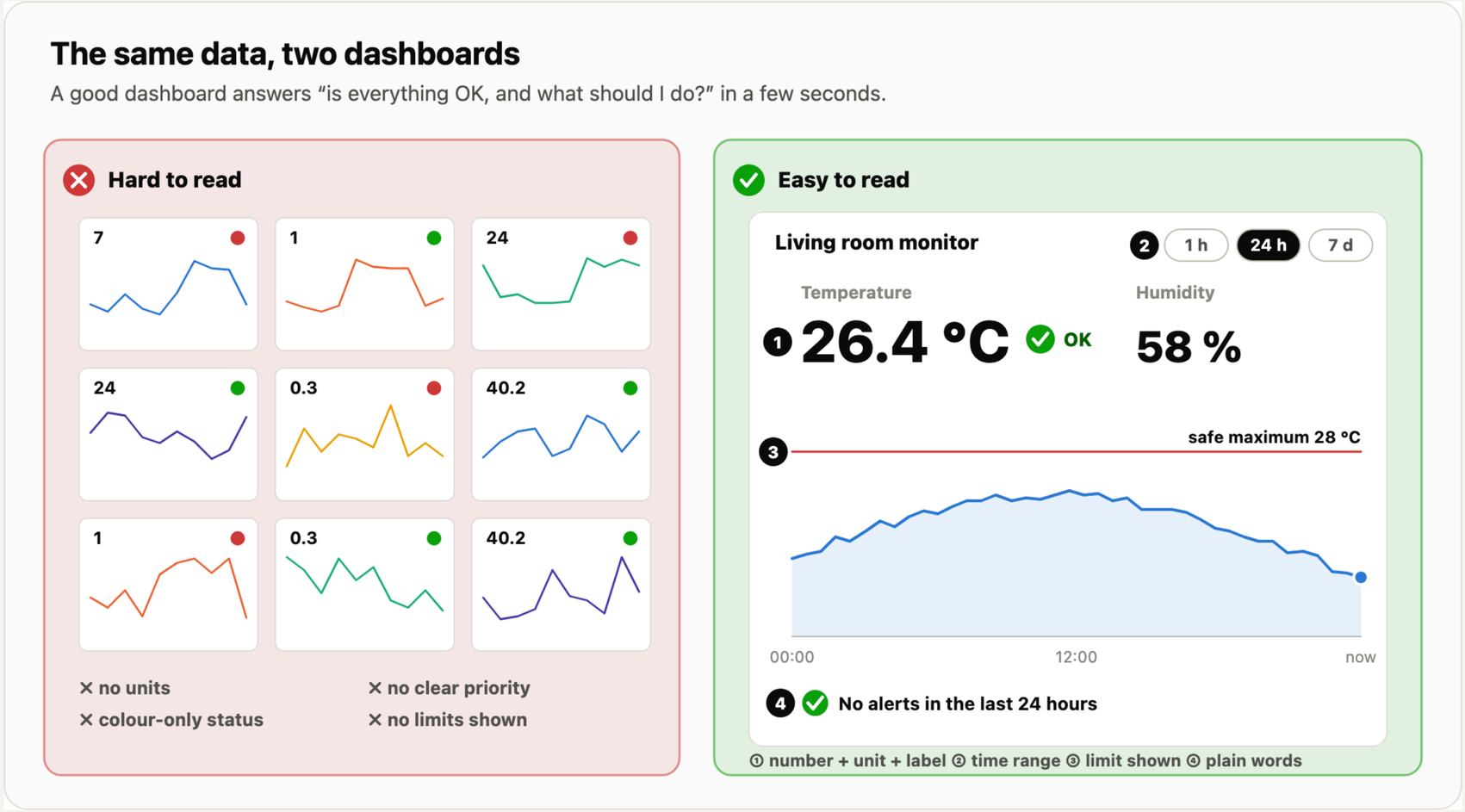
Different questions need different visuals

To show…	Use a…	Example
Change over time	Line chart	Shelter temperature, last 24 hours
Value right now	Gauge or big number	“Fuel tank is 72% full”
A comparison	Bar chart	Diesel used by each site this month
On/off, good/bad	Status indicator	“Generator: RUNNING”
Where things are	Map	Tower sites in the region
Exact values	Table	The last 10 readings, with times

For IoT sensor data, the **line chart** is the workhorse.

DASHBOARD DESIGN

Good dashboard design in action



PAUSE AND CHECK

A hospital wants one screen showing whether each of 20 medicine fridges is in its safe range. Which visual would you choose, and what would you avoid?

Think of the NOC view. Discuss in your team for 30 seconds.

Part C

Energy: the view managers need

Diesel, power and cooling are some of the biggest bills in the network.

ENERGY MANAGEMENT

The energy view of the tower network

Diesel

Litres used per site, per month

Energy mix

Grid, battery, generator and solar, per site

Run-hours

Generator hours, compared with the fuel level

Hybrid sites

EDOTCO: hybrid renewable power for off-grid Sabah sites

Energy is **15–20%** of a mobile operator’s operating costs. Sources: GSMA Intelligence, 2025 · EDOTCO, 2019

ENERGY IN OUR OWN DATA CENTRES

PUE: how efficient is the data centre?

1.56

average PUE in 2024, flat for five years

>30%

of power on cooling in less efficient sites

PUE = all the site's electricity ÷ the IT equipment's electricity. **1.56:** for every 1 kWh of IT, another 0.56 kWh goes on cooling and losses.

Source: Uptime Institute Global Data Center Survey 2024 (PUE) · IEA, *Energy and AI*, 2025 (cooling)

THE BUSINESS LENS

Which three numbers would you show the board about the tower network?

Think about money, risk and customers. Start now, and decide in the lab.

IN TELCO OPERATIONS

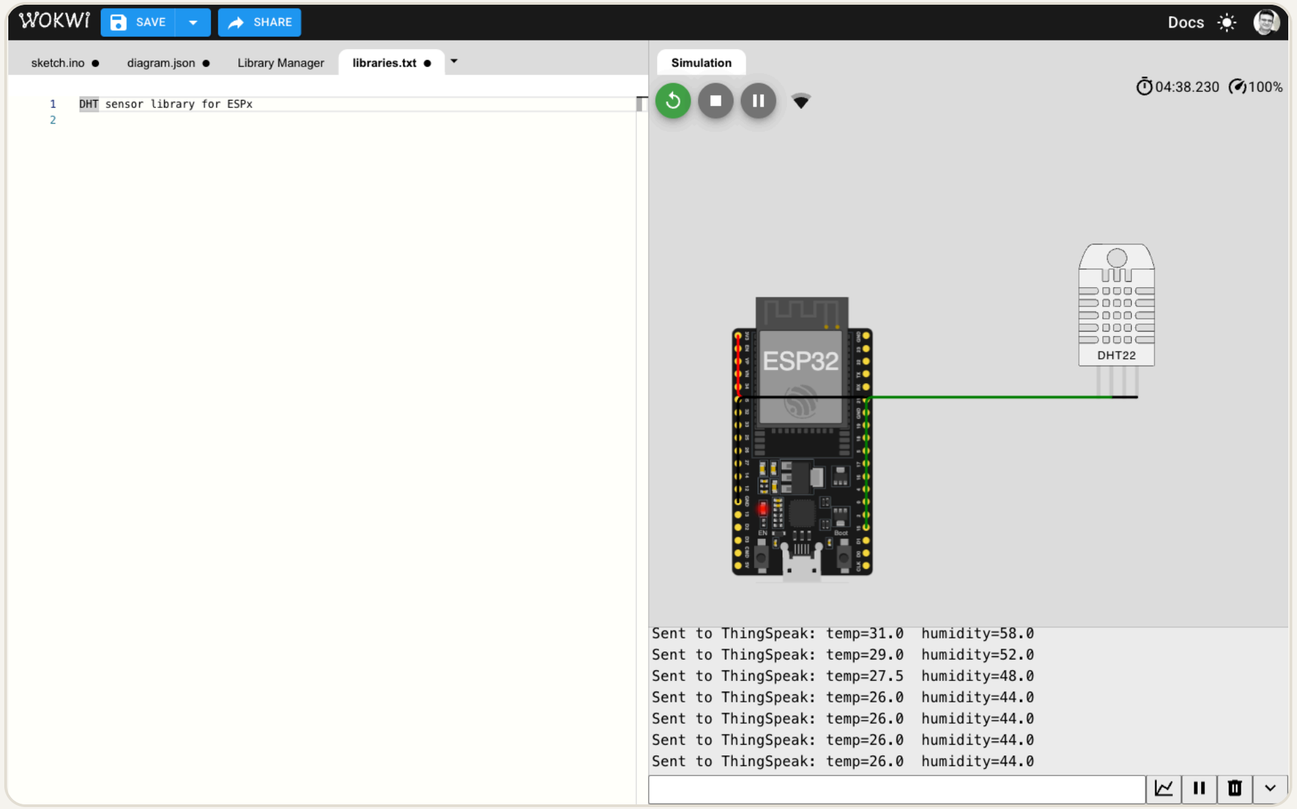
Cloud dashboards in telco operations

	Today	With IoT	The benefit
NOC alarms	Many alarms: which matter?	A status light per site	Faster fault repair
Field visits	An engineer drives out to look	Check the trend; go if needed	Fewer site visits
Site energy	Diesel seen on visits and bills	Diesel and energy mix per site	Less diesel, less fuel theft

Read each row: **today → with IoT → the benefit**

LAB 4 · 22 MINUTES HANDS-ON

Lab 4: the trend view of one tower site



1. Create a ThingSpeak channel with two fields
2. Copy your **Write API key**
3. Paste the key into Wokwi and press Play
4. Watch the charts grow every 20 seconds
5. **Manager task:** 3 numbers for a board report

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- An **IoT platform** receives, stores, analyses and shows site data
- **ThingSpeak:** channels, fields, and a Write key you keep secret
- **The NOC view:** a status light per site, worst problems first
- **The energy view:** diesel, energy mix and run-hours per site
- **Line charts** for change over time; status lights for “right now”
- **PUE 1.56:** in data centres, 0.56 kWh extra for every 1 kWh of IT

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 The NOC watches 500 tower sites. What should its screen show first?

Q2 Name two things an energy view shows for each tower site.

Q3 Which chart shows how a shelter’s temperature changed last week?

Q4 In ThingSpeak, what does the Write key do? And the Read key?

UP NEXT · MODULE 6

Data Processing and Automation

The Smart Site Monitor starts to act: extra cooling in the shelter switches on at 28 °C and off at 26 °C, with no cloud needed.

Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop